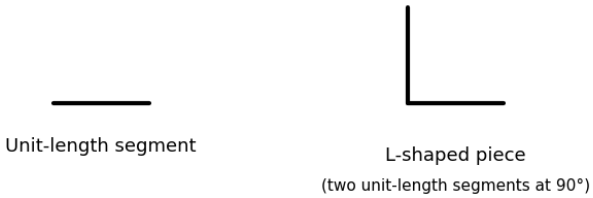
A. Grid L

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: brute force, constructive algorithms, math, number theory | https://codeforces.com/problemset/problem/2219/A

Roger has p unit-length segments and q L-shaped pieces, each formed by joining two unit-length segments at a right angle.



He wants to use all of these pieces, with no piece left unused, to form a grid of dimensions $n \times m$.

Given p and q, determine whether there exist positive integers n and m such that an $n \times m$ grid can be formed using exactly p unit-length segments and q L-shaped pieces, possibly rotating them, and provide any valid n and m.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 100$). The description of the test cases follows.

The first line of each test case contains two integers p and q ($1 \le p, q \le 10^8$).

Output

For each test case, print any valid n and m such that it's possible to construct $n \times m$ using all pieces. If there are no such n and m, print -1.

Example

Input
7 1 2 1 3 5 1 2 5 2 10 100000000 100000000 1 1

Output

```
-1
1 2
1 2
2 2
2 4
-1
-1
```

Note

Here are the constructions for examples 2, 3, and 4. We use a different color for different L-shaped pieces, black for all unit-length segments.

